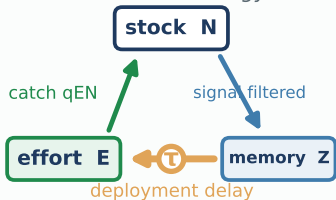


THE DELAY SITS IN GOVERNANCE

not in the ecology



the delay is institutional

assessment → decision → action

not an ecological lag

mobilising law

gain grows with
deployment

protective law

restores toward
a quota cap

TWO CHANNELS, OPPOSITE MATH

stability vs the deployment delay τ

subcritical Hopf pair — interval-certified
3.7 150

mobilising



*the lag, not the feedback,
holds the window stable*

protective

stable at every τ

τ (yr, log)

no-Hopf theorem at every τ

loop gain peaks at $0.080 < 1$

continuation: a five-regime attractor topology

THE REVIEW INTERVAL AS CONTROL

sample-and-hold governance

Neimark-Sacker-type crossing (exact update)

6.5

mobilising



Euler 47.5 — artefact

protective

annual review: stable

review interval T_r (yr, log)

the form and timing of response,

not ecological lag alone, decide
whether governance stabilises

Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter